

Healthcare SQL Project

Database: healthcare_db | Table: healthcare_dataset | Rows: 1000

Dataset Columns

Column	Type	Example
Name	VARCHAR	Bobby Jackson
Age	INT	30
Gender	VARCHAR	Male
Blood Type	VARCHAR	B-
Medical Condition	VARCHAR	Cancer
Date of Admission	DATE	2024-01-31
Doctor	VARCHAR	Dr. Matthew Smith
Hospital	VARCHAR	Sons and Miller
Insurance Provider	VARCHAR	Blue Cross
Billing Amount	DECIMAL	18856.28
Room Number	INT	328
Admission Type	VARCHAR	Urgent
Discharge Date	DATE	2024-02-02
Medication	VARCHAR	Paracetamol
Test Results	VARCHAR	Normal

WHERE Queries

Q1: Find all Female patients

```
SELECT * FROM healthcare_dataset
WHERE gender = 'Female';
```

✓ Output: 476 rows

Q2: Find all patients with Cancer

```
SELECT * FROM healthcare_dataset
WHERE `Medical Condition` = 'Cancer';
```

✓ Output: 147 rows

Q3: Find patients with Billing Amount > 40000

```
SELECT * FROM healthcare_dataset
WHERE `Billing Amount` > 40000;
```

✓ Output: 209 rows

Q4: Find all Emergency admissions

```
SELECT * FROM healthcare_dataset
WHERE `Admission Type` = 'Emergency';
```

✓ Output: 309 rows

Q5: Find Male patients with Abnormal test results

```
SELECT * FROM healthcare_dataset
WHERE `Test Results` = 'Abnormal'
AND Gender = 'Male';
```

✓ Output: 189 rows

GROUP BY + ORDER BY Queries

Q6: Count patients per Medical Condition

```
SELECT `Medical Condition`, COUNT(*) AS total_patients
FROM healthcare_dataset
GROUP BY `Medical Condition`
ORDER BY total_patients;
```

✓ Output: Hypertension-192, Arthritis-183, Asthma-175, Obesity-162, Cancer-147, Diabetes-141

Q7: Total billing amount per Hospital

```
SELECT Hospital, SUM(`Billing Amount`) AS total_amount
FROM healthcare_dataset
GROUP BY Hospital
ORDER BY total_amount;
```

✓ Output: Medanta Hospital - 32,90,487 (Highest)

Q8: Average billing per Insurance Provider

```
SELECT `Insurance Provider`, AVG(`Billing Amount`) AS Average_billing
FROM healthcare_dataset
GROUP BY `Insurance Provider`
ORDER BY Average_billing DESC;
```

✓ Output: Medicare - 28,030 (Highest avg)

Q9: Count of each Admission Type

```
SELECT `Admission Type`, COUNT(*) AS total_Admission
FROM healthcare_dataset
GROUP BY `Admission Type`
ORDER BY total_Admission;
```

✓ Output: Elective-356, Urgent-335, Emergency-309

Q10: Most used Medications

```
SELECT Medication, COUNT(*) AS usage_count
FROM healthcare_dataset
GROUP BY Medication
ORDER BY usage_count DESC;
```

✓ Output: Omeprazole-112 (Most used)

String Function Queries

Q11: Display all patient names in UPPERCASE

```
SELECT UPPER(Name) AS patient_name
FROM healthcare_dataset;
```

✓ Output: All names in uppercase

Q12: Find patients whose name starts with 'A'

```
SELECT * FROM healthcare_dataset
WHERE Name LIKE 'A%';
```

✓ Output: 179 rows

Q13: Extract first name only from full name

```
SELECT SUBSTRING_INDEX(Name, ' ', 1) AS first_name
FROM healthcare_dataset;
```

✓ Output: First names of all patients

Q14: Find conditions containing 'ia'

```
SELECT Name, `Medical Condition`
FROM healthcare_dataset
WHERE `Medical Condition` LIKE '%ia%';
```

✓ Output: 141 rows (Diabetes patients)

Q15: Combine Doctor and Hospital in one column

```
SELECT CONCAT(Doctor, ' @ ', Hospital) AS doctor_info
FROM healthcare_dataset;
```

✓ Output: Dr. Rajesh Kumar @ City Hospital

Mixed Real World Queries

Q16: Top 5 highest billed patients

```
SELECT Name, `Medical Condition`, `Billing Amount`
FROM healthcare_dataset
ORDER BY `Billing Amount` DESC
LIMIT 5;
```

✓ Output: Leslie Terry - 49,995 (Highest)

Q17: Count of Male patients above 60 with Abnormal results

```
SELECT COUNT(*) FROM healthcare_dataset
WHERE Gender = 'Male'
AND Age > 60
AND `Test Results` = 'Abnormal';
```

✓ Output: 64

Q18: Medical Conditions with more than 150 cases

```
SELECT `Medical Condition`, COUNT(*) AS Total_cases
FROM healthcare_dataset
GROUP BY `Medical Condition`
HAVING COUNT(*) > 150;
```

✓ Output: Hypertension, Arthritis, Asthma, Obesity

Q19: Doctor with highest total billing

```
SELECT Doctor, SUM(`Billing Amount`) AS total_amount
FROM healthcare_dataset
GROUP BY Doctor
ORDER BY total_amount DESC
LIMIT 1;
```

✓ Output: Dr. Priya Sharma - 32,97,759

Q20: Average age per Medical Condition

```
SELECT `Medical Condition`, AVG(Age) AS Average_age
FROM healthcare_dataset
GROUP BY `Medical Condition`;
```

✓ Output: Cancer - 54.38 (Highest avg age)

Project Summary

Category	Questions	Status
WHERE	5	✓ Complete
GROUP BY + ORDER BY	5	✓ Complete
String Functions	5	✓ Complete
Mixed Real World	5	✓ Complete
TOTAL	20	✓ All Done